

The aim of the work was to implement a Python program for the gathering, analysis, and visualization of meteorological data using the Meteo.lt REST API and the Pandas library.

Step 1:

Class WeatherAPI was implemented, it contains 2 methods – get_historical_data and forecast_data, utilizing pd.DatetimeIndex for time zone management.

Historical data: collected via {self.url_address}/stations/{self.location}/observations/{data_str} for period September 12, 2024 to September 12, 2025. Data was parsed and structured into Pandas DataFrame.

Forecast data: obtained using {self.url_address}/places/{place}/forecasts/long-term”.

Step 2

Based on the annual historical dataset, following analytical metrics were computed:

Average temperature 8.60979052823315

Average humidity 79.01229508196721

Average temperature during the day 10.04148451730419

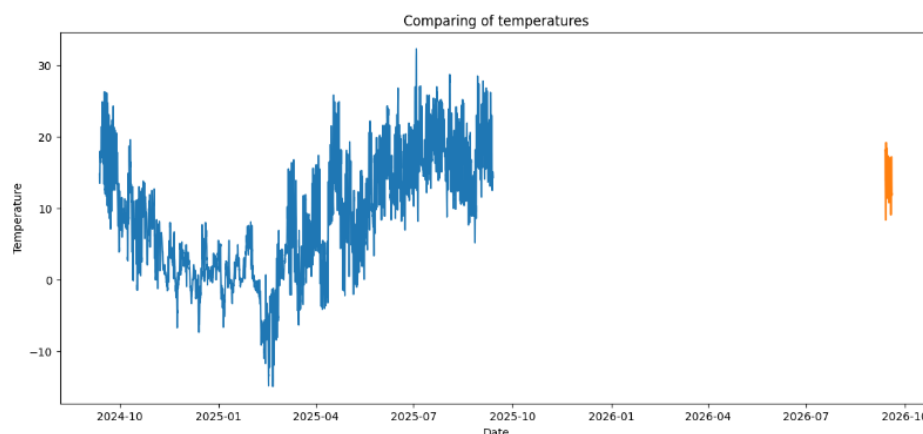
Average temperature during the night 7.178096539162113

Rainy weekend {datetime.date(2024, 10, 13), datetime.date(2024, 9, 28), datetime.date(2025, 5, 24), datetime.date(2025, 7, 20), datetime.date(2025, 5, 4), datetime.date(2025, 3, 2), datetime.date(2025, 8, 16), datetime.date(2025, 5, 17), datetime.date(2025, 5, 11), datetime.date(2025, 7, 27), datetime.date(2025, 1, 26), datetime.date(2025, 2, 16), datetime.date(2025, 5, 10), datetime.date(2025, 6, 8), datetime.date(2025, 8, 23), datetime.date(2025, 8, 31), datetime.date(2024, 11, 16), datetime.date(2025, 6, 29), datetime.date(2025, 3, 16), datetime.date(2024, 10, 27), datetime.date(2024, 12, 22), datetime.date(2025, 8, 24), datetime.date(2024, 12, 8), datetime.date(2025, 5, 18), datetime.date(2025, 8, 17), datetime.date(2024, 12, 21), datetime.date(2025, 7, 19), datetime.date(2024, 10, 6), datetime.date(2025, 7, 26), datetime.date(2024, 12, 15), datetime.date(2025, 2, 1), datetime.date(2025, 8, 10), datetime.date(2024, 12, 28), datetime.date(2025, 6, 22), datetime.date(2025, 2, 2), datetime.date(2025, 2, 15), datetime.date(2025, 1, 18), datetime.date(2024, 9, 15), datetime.date(2025, 5, 3), datetime.date(2024, 11, 17), datetime.date(2025, 8, 2), datetime.date(2025, 1, 25), datetime.date(2024, 11, 2), datetime.date(2025, 3, 23), datetime.date(2024, 11, 30), datetime.date(2025, 4, 5), datetime.date(2025, 4, 19), datetime.date(2025, 7, 13), datetime.date(2025, 1, 4), datetime.date(2025, 3, 1), datetime.date(2024, 11, 23), datetime.date(2025, 1, 5), datetime.date(2025, 1, 12), datetime.date(2025, 6, 7), datetime.date(2024, 11, 3), datetime.date(2025, 5, 31), datetime.date(2025, 7, 6)}

Number of rainy weekends 57

The results show a logical meteorological pattern, where average nighttime temperature is lower than daytime temperature. Over the year, 57 weekend periods precipitation events.

Step 3



Blue line shows historical data, demonstrating seasonal fluctuations from winter negative temperatures to summer heat waves. Orange line represents short prediction for close period that was received from long-term forecast API. Gap between lines happened because of periods.

Step 4

It was written in one line. Principle of work of `resample('5min')` – it changes discretization of time row, creating span with step 5 minutes instead of initial time intervals. For new artificially created points values initially fill `Nan`. Role of linear interpolation - fills in the gaps created by drawing a straight line between adjacent hourly measurements. It evenly distributes the temperature difference between two hours into 12 intermediate five-minute increments.